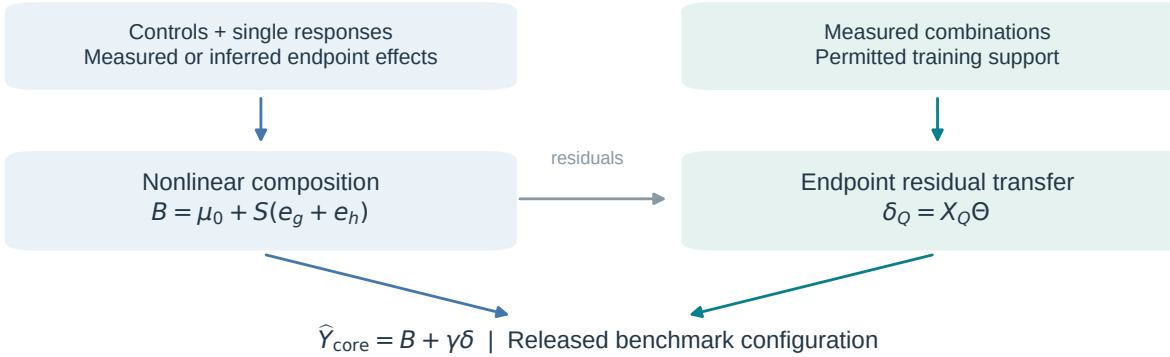
a **Predict responses from measured interventions**



b **Remove endpoint contributions**

A+B A+D C+B C+D

$$\Delta = Y_{AB} - Y_{AD} - Y_{CB} + Y_{CD}$$

$$P_Q = I - X_Q X_Q^{+}$$

Partner contrasts remain after
separate endpoint contributions cancel

c **Learn additional partner responses**

RNA: projected pair kernel
Training-half readout weights

Function: indexed residual matrix
Rank one / spectral correction

Function: shared SF12 / V4
Single effects + gene annotations

d **Test the response component changed by each branch**

Added branch	Complete response	Partner contrast
Endpoint residual	Can change	Unchanged: $P_Q\delta_Q = 0$
Explicit partner learner	Can change	Can change; evaluate its gain

e **Adjust the partner correction while preserving endpoints**

$$\widehat{Y}_\alpha = M_0 + (I - P_Q)D + \alpha P_Q D$$

Retain endpoint part | Adjust partner part

Internal held-combination predictions set $\alpha \in [0, 1]$

Specified query panel | Query identities used; query outcomes excluded